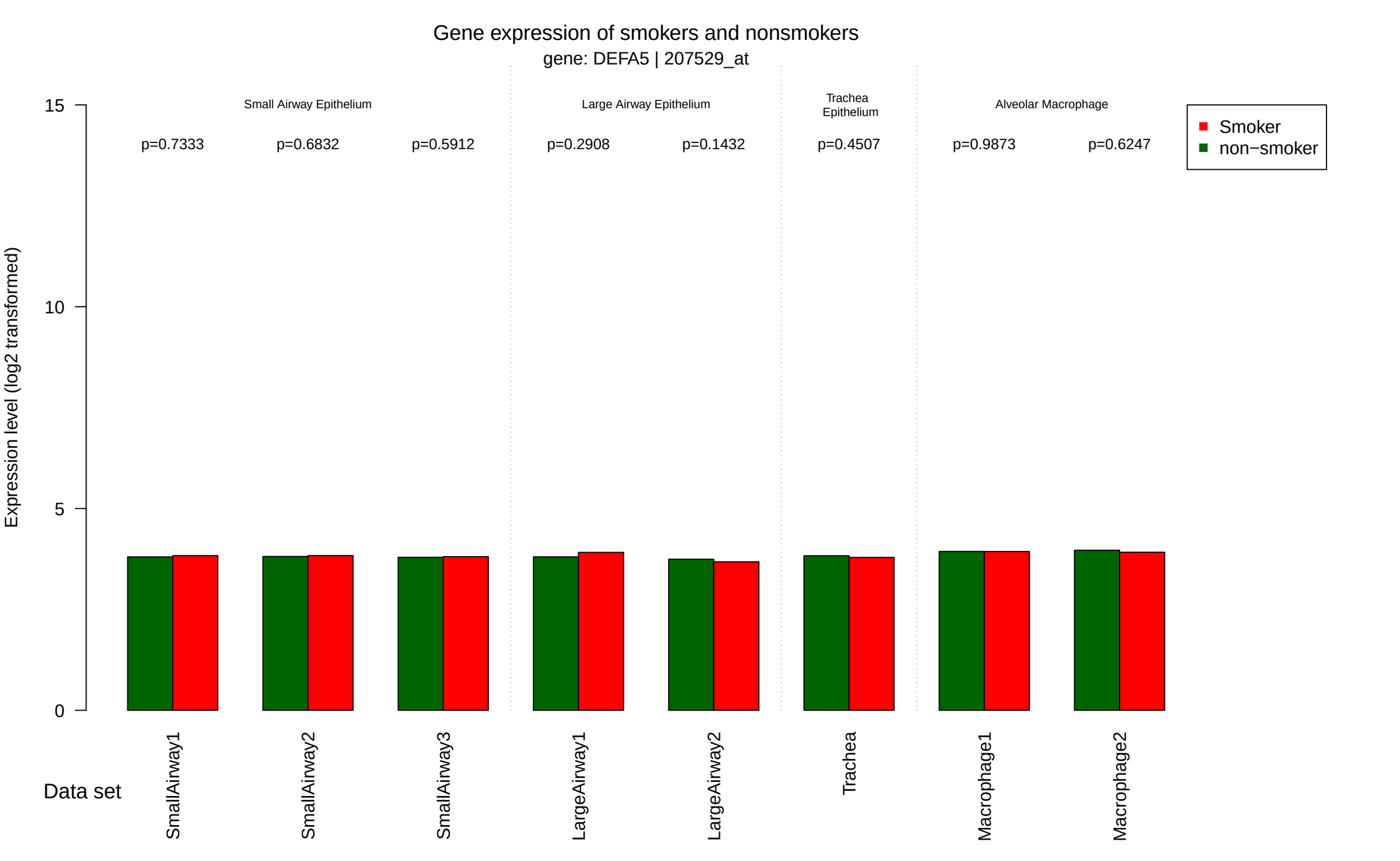
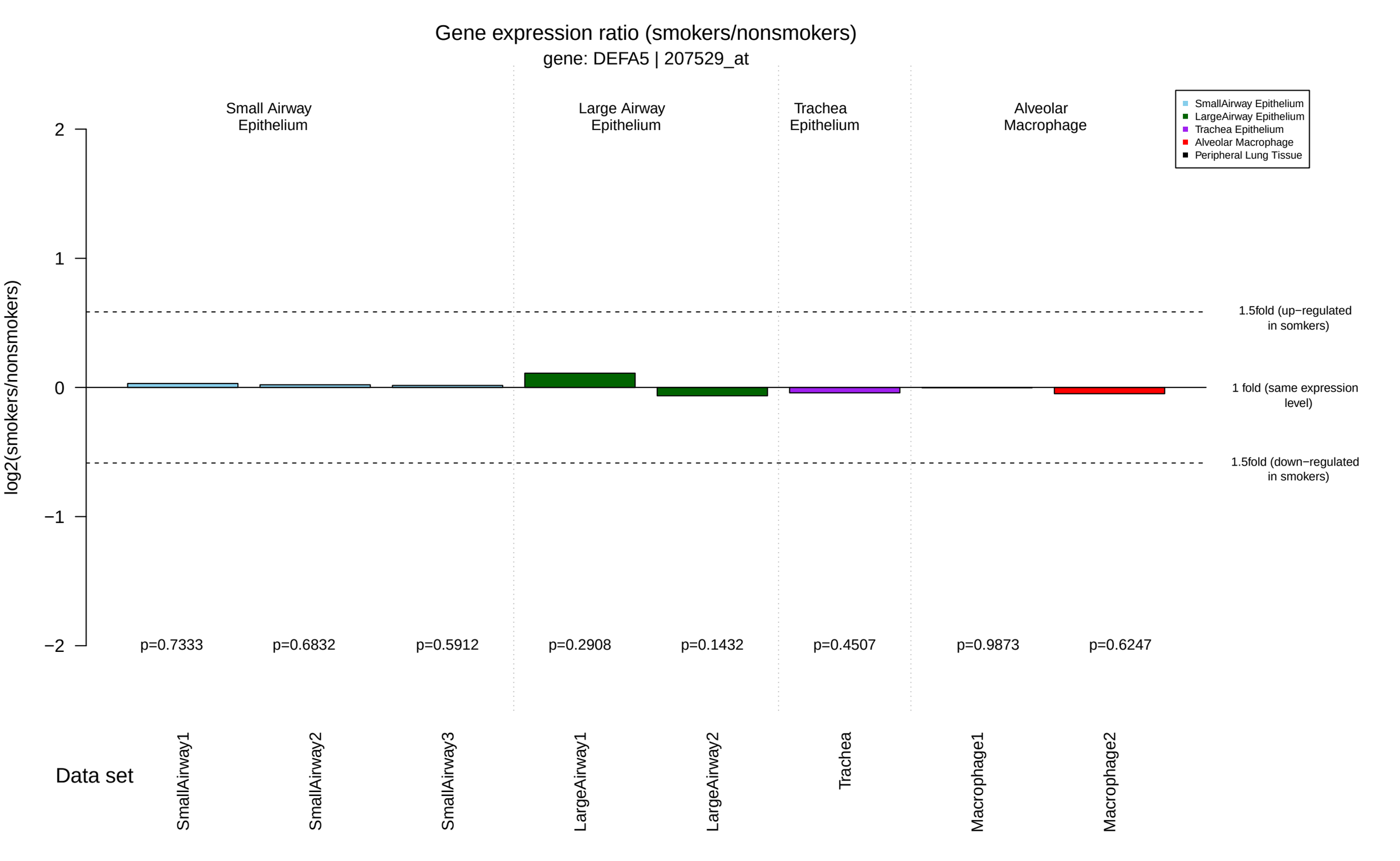
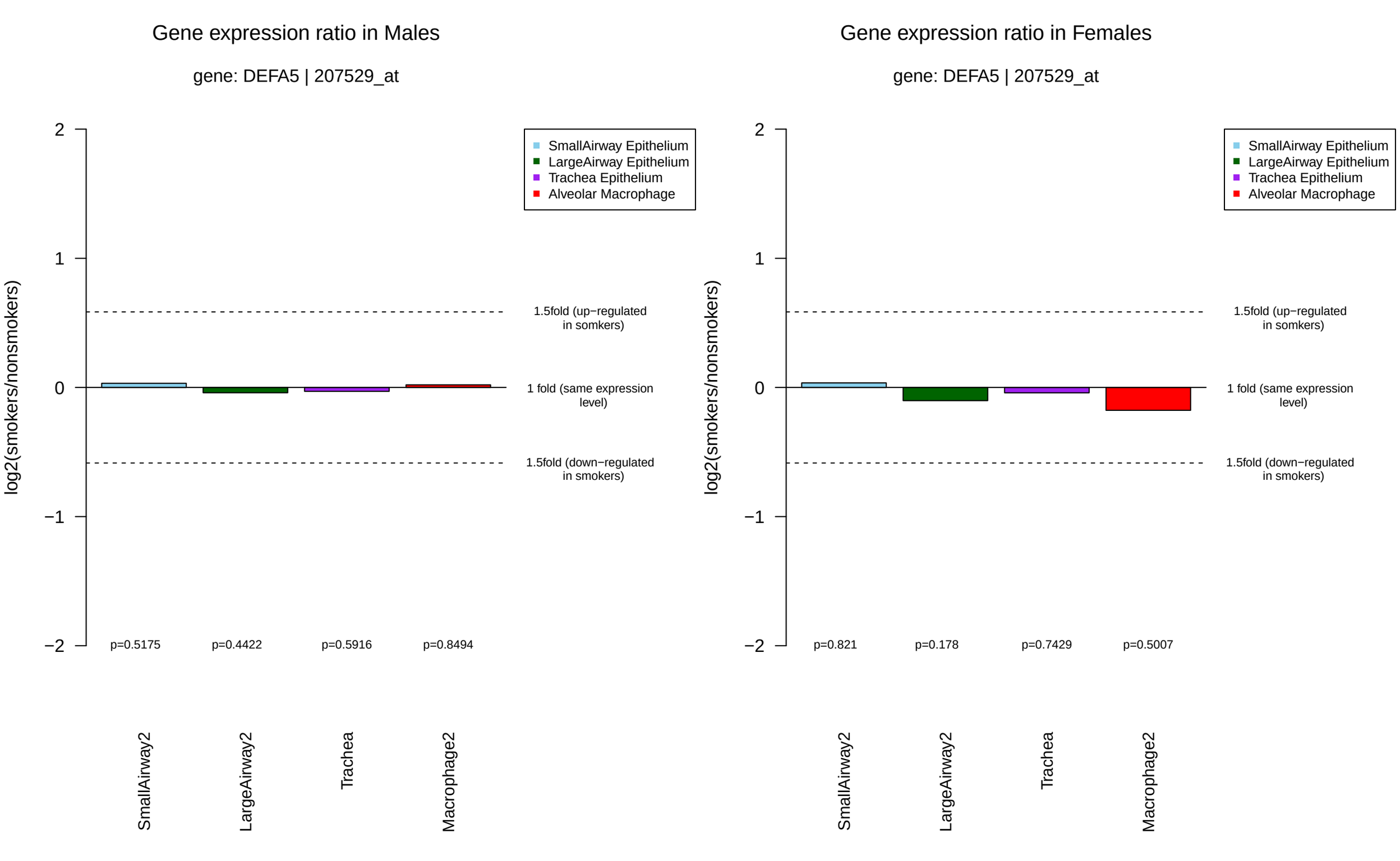
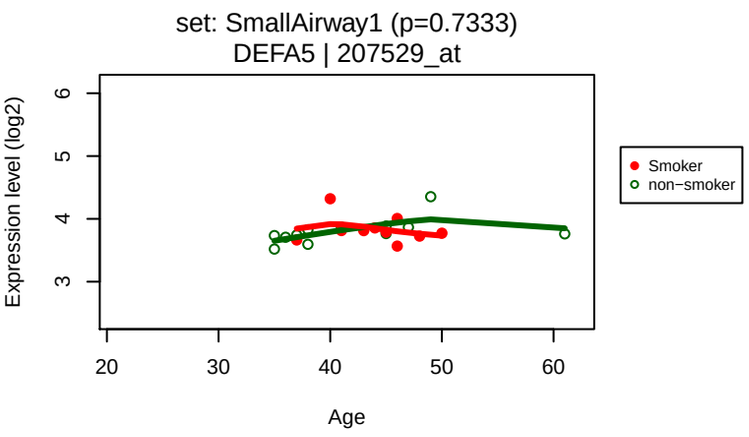
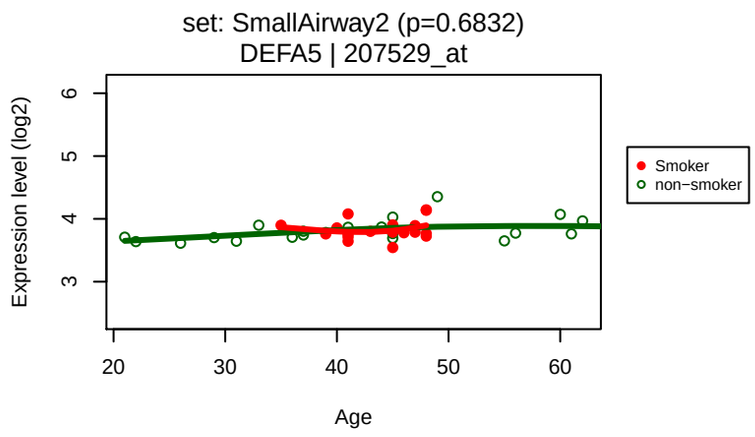
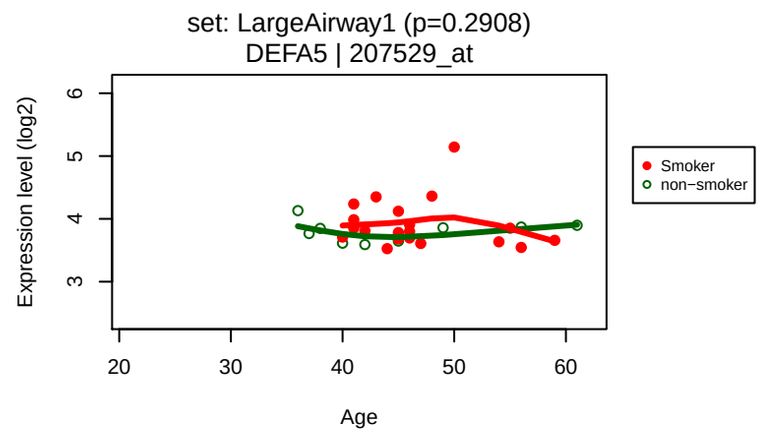

Gene expression of smokers and nonsmokers
gene: DEFA5 | 207529_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7333
p=0.6832
p=0.5912
p=0.2908
p=0.1432
p=0.4507
p=0.9873
p=0.6247
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

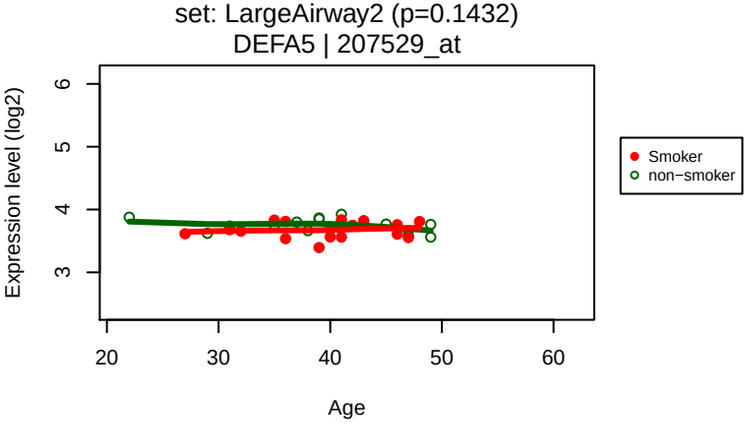

Gene expression ratio (smokers/nonsmokers)
gene: DEFA5 | 207529_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7333
p=0.6832
p=0.5912
p=0.2908
p=0.1432
p=0.4507
p=0.9873
p=0.6247
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

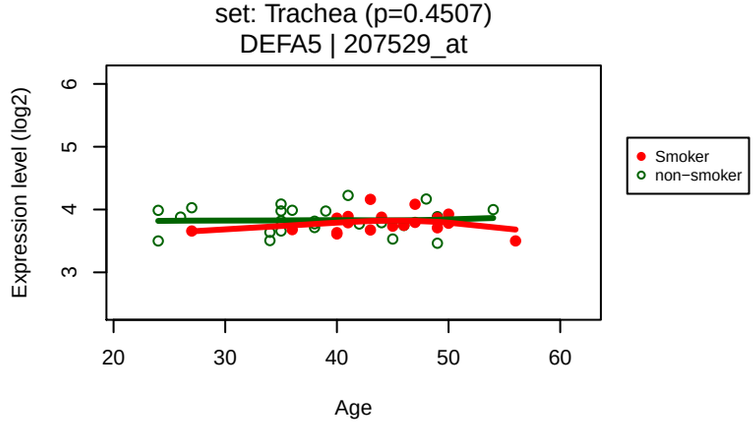

Gene expression ratio in Males
gene: DEFA5 | 207529_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5175
p=0.4422
p=0.5916
p=0.8494
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: DEFA5 | 207529_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.821
p=0.178
p=0.7429
p=0.5007
SmallAirway2
LargeAirway2
Trachea
Macrophage2

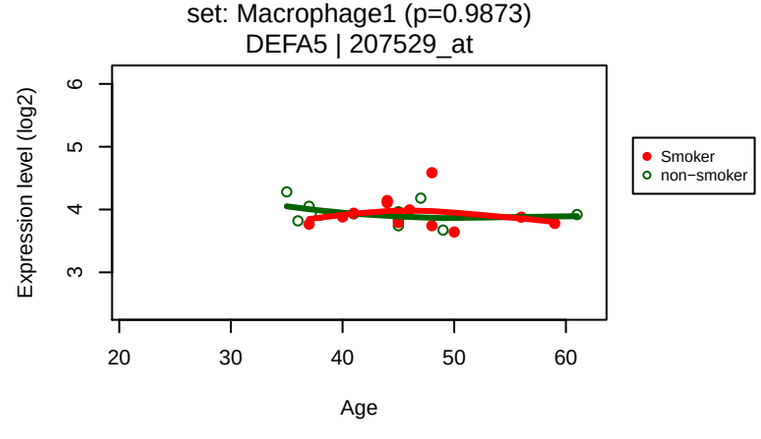

set: SmallAirway1 (p=0.7333)
DEFA5 | 207529_at
Expression level (log2)
Age
Smoker
non−smoker

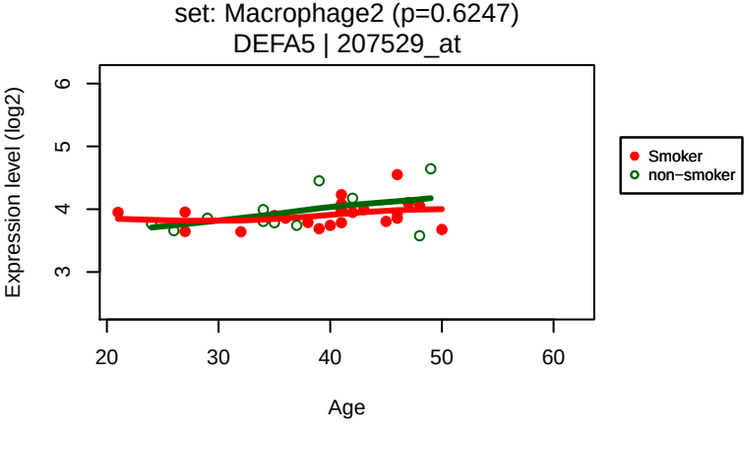

set: SmallAirway2 (p=0.6832)
DEFA5 | 207529_at
Expression level (log2)
Age
Smoker
non−smoker

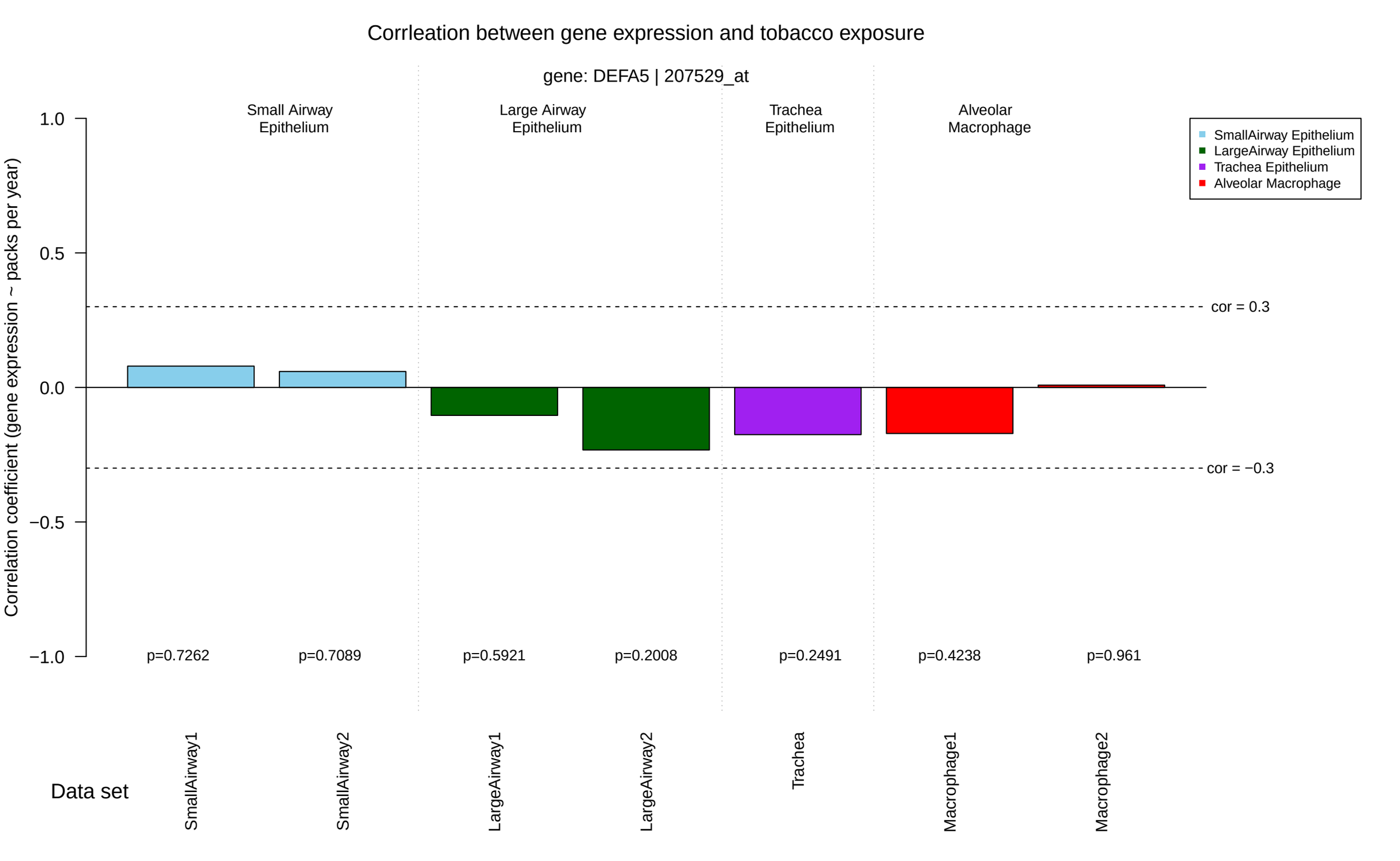

set: LargeAirway1 (p=0.2908)
DEFA5 | 207529_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1432)
DEFA5 | 207529_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4507)
DEFA5 | 207529_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9873)
DEFA5 | 207529_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6247)
DEFA5 | 207529_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: DEFA5 | 207529_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7262
p=0.7089
p=0.5921
p=0.2008
p=0.2491
p=0.4238
p=0.961
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2